Fine-mapped single-cell eQTL evaluation of AlphaGenome fine-tuning

1 Overview

We evaluate whether sequence-derived mutation effects distinguish fine-mapped single-cell expression quantitative trait loci (eQTLs). A positive is a variant with posterior inclusion probability (PIP) greater than 0.75 and a negative is a variant with PIP less than 0.01. The evaluation uses human fine-mapping panels and a 1 Mb sequence window centered on the transcription start site (TSS) of the associated gene. RNA-seq denotes RNA sequencing and CAGE denotes Cap Analysis of Gene Expression. Average precision (AP) is reported as area under the precision-recall curve (AUPR); AUROC denotes area under the receiver operating characteristic curve.

The primary comparison is a trained head-only model versus three joint fine-tuned models. Two share the head-only model's targets and add low-rank adaptation (LoRA) or LoRA with convolutional low-rank adapters (LoCon). The third is the LoRA+LoCon model with the best held-out correlation in the joint fine-tuning comparison, which differs from the others only in using Zemke 2024 ATAC targets rebuilt from raw fragments, 18 broad groups in place of 22 released tracks (raw-Zemke). The benchmark retains up to 1,000 variants in each PIP class per fine-mapping panel with a fixed random seed. The raw and retained counts are recorded with the benchmark output.

2 Cell-type matching

The 35 fine-mapping panels use the prefixes Ast (astrocyte), End (endothelial), Ext (excitatory), IN (inhibitory), MG (microglia), OD (oligodendrocyte), and OPC (oligodendrocyte precursor cell), with numbered panels denoting subpanels. All four trained models have the same 20 Zemke 2023 RNA output tracks, and track positions are looked up by label in each loaded head. The matching is shown below. The L5 IT output is excluded because its RNA training target was invalid, so that channel received no supervision. A separate native AlphaGenome analysis evaluates all valid pretrained RNA-seq and CAGE tracks, without adding a newly initialized head.

Table 1: Fine-mapping panel to RNA-track matching for all trained models. A numbered panel inherits the mapping for its prefix.

Panel prefix	Zemke 2023 RNA tracks
`Ast*`	ASC
`End*`	Endo
`Ext*`	Mean of L2/3 IT, L4/5 IT, L5/6 NP, L5 ET, L6 CT, L6 IT, L6 IT CAR3, and L6b
`IN*`	Mean of LAMP5, PVALB, SNCG, SST, and VIP
`MG*`	MGC
`OD*`	ODC
`OPC*`	OPC

The panel labels do not exactly match the available pretrained output tracks. The native model therefore receives a separate track-level analysis; it is not represented by the trained-head comparison above.

3 Mutation effects

For variant i, let $\mathbf{y}_{i,r} \in \mathbb{R}^{B\times C}$ and $\mathbf{y}_{i,a} \in \mathbb{R}^{B\times C}$ be the reference and alternate RNA predictions at the model's 128 base-pair (bp) resolution, where B is the number of output bins and $C = 19$ is the number of RNA tracks. Let $\mathcal{C}_g$ be the track indices for the matched cell group. The TSS score is

$$s_i^{\mathrm{TSS}} = \frac{1}{|\mathcal{C}_g|} \sum_{c\in\mathcal{C}_g} \left| y_{i,a,b_{\mathrm{TSS}},c} - y_{i,r,b_{\mathrm{TSS}},c} \right|, \tag{1}$$

where b_{TSS} is the 128 bp bin containing the TSS. The gene score is the corresponding mean absolute alternate-reference difference after averaging each track over all 128 bp bins overlapping the annotated gene span,

$$s_i^{\mathrm{gene}} = \frac{1}{|\mathcal{C}_g|} \sum_{c\in\mathcal{C}_g} \left| \frac{1}{|\mathcal{B}_i|} \sum_{b\in\mathcal{B}_i} y_{i,a,b,c} - \frac{1}{|\mathcal{B}_i|} \sum_{b\in\mathcal{B}_i} y_{i,r,b,c} \right|, \tag{2}$$

where $\mathcal{B}_i$ is the set of bins overlapping the gene span. Both scores are nonnegative because fine-mapping identifies variant probability, not the direction of expression change. Performance is measured by AUROC and average precision, overall and within bins of |variant − TSS|.

4 Results

Each trained model scored the same 54,193 variants after reference-allele validation. An earlier version of this benchmark read the Zemke 2023 RNA head with a 19-label track list that omitted L6 IT, which shifted every track from position 8 onward. Microglia, oligodendrocyte, OPC, and inhibitory panels were therefore scored with other cell types' tracks. All results below use the corrected label lookup.

Table 2 reports AUROC and AP pooled over all panels, and AUROC averaged over the seven panel groups after computing it within each group. Pooled metrics rank variants from different panels against one another, so they also depend on the scale of each group's track effects; the within-group mean depends only on the ranking inside each panel. The raw-Zemke LoRA+LoCon model has the highest pooled TSS-bin and gene-span AUROC and AP. The head-only model has the highest within-group gene-span AUROC. Differences between trained models are at most 0.016 in AUROC, comparable to the variation of about 0.005 observed between repeated runs of the same model with nondeterministic GPU kernels. The native matched tracks are listed for reference; they were selected on the same variants and may be optimistic.

Performance by distance from the TSS is shown below. Per-bin AUROC, AP, and class counts are available in the machine-readable summary output.

4.1 Trained track specificity

If the trained RNA tracks carry cell-type-specific signal, each panel should be discriminated best by its own group's tracks. For each panel group p and track group q, let $A_{p,q}$ be the AUROC of panel p

Table 2: Overall eQTL discrimination across scored variants. AP is average precision. "Within" is the mean of per-group AUROC over the seven panel groups.

	TSS bin			Gene span		
Model	AUROC	AP	Within	AUROC	AP	Within
Head-only	0.512	0.391	0.513	0.547	0.444	0.550
LoRA	0.521	0.404	0.524	0.540	0.432	0.539
LoRA+LoCon	0.517	0.399	0.518	0.540	0.434	0.542
LoRA+LoCon, raw Zemke 2024 ATAC	0.523	0.421	0.527	0.547	0.445	0.545
Native matched tracks	0.500	0.367	0.526	0.524	0.439	0.535

variants scored with the composite effect of group q, the mean effect over that group's tracks. With K groups, the specificity statistic is

$$D = \frac{1}{K}\sum_{p} A_{p,p} - \frac{1}{K(K-1)}\sum_{p\neq q} A_{p,q}, \tag{3}$$

the mean matched AUROC minus the mean unmatched AUROC. Its p-value is the fraction of the $K!$ assignments of composites to panels whose statistic is at least D; with $K = 7$ the smallest attainable value is $1/5040$.

Table 3 shows that D is positive for every model and summary, with $p \leq 0.006$. The size of the effect is small, 0.015 to 0.032 in AUROC. Figure 3 and the rank output show that the matched composite ranks first for microglia panels in every model and summary, and for astrocyte panels in six of eight model and summary combinations. The matched composite ranks first for oligodendrocyte panels in five and for OPC panels in two of the eight combinations. Excitatory and inhibitory panels mostly rank their own composite second to fourth; these composites average closely related neuronal tracks. The endothelial composite is the weakest predictor for most other panels. The median Spearman correlation between the effects of different tracks on the same variants is 0.63 to 0.72, so most of each variant's predicted effect is shared across cell types.

Table 3: Cell-type specificity of trained RNA tracks. D is the matched minus unmatched mean AUROC and p its exact permutation p-value; ρ is the median Spearman correlation between track effects.

	TSS bin			Gene span		
Model	D	p	ρ	D	p	ρ
Head-only	0.016	0.0020	0.69	0.025	0.0010	0.63
LoRA	0.021	0.0010	0.72	0.024	0.0006	0.66
LoRA+LoCon	0.018	0.0012	0.71	0.032	0.0002	0.66
LoRA+LoCon, raw Zemke 2024 ATAC	0.015	0.0060	0.72	0.025	0.0002	0.68

4.2 Native track specificity

The native AlphaGenome evaluation stores signed alternate-minus-reference effects for 54,729 sampled variants and 1,213 valid pretrained channels, 667 RNA-seq and 546 CAGE. Predictions are reduced on device to a 128 bp TSS bin and the mean over the annotated gene span. Of the variants, 54,193 have finite effects and 536 were skipped because the reference allele did not match the

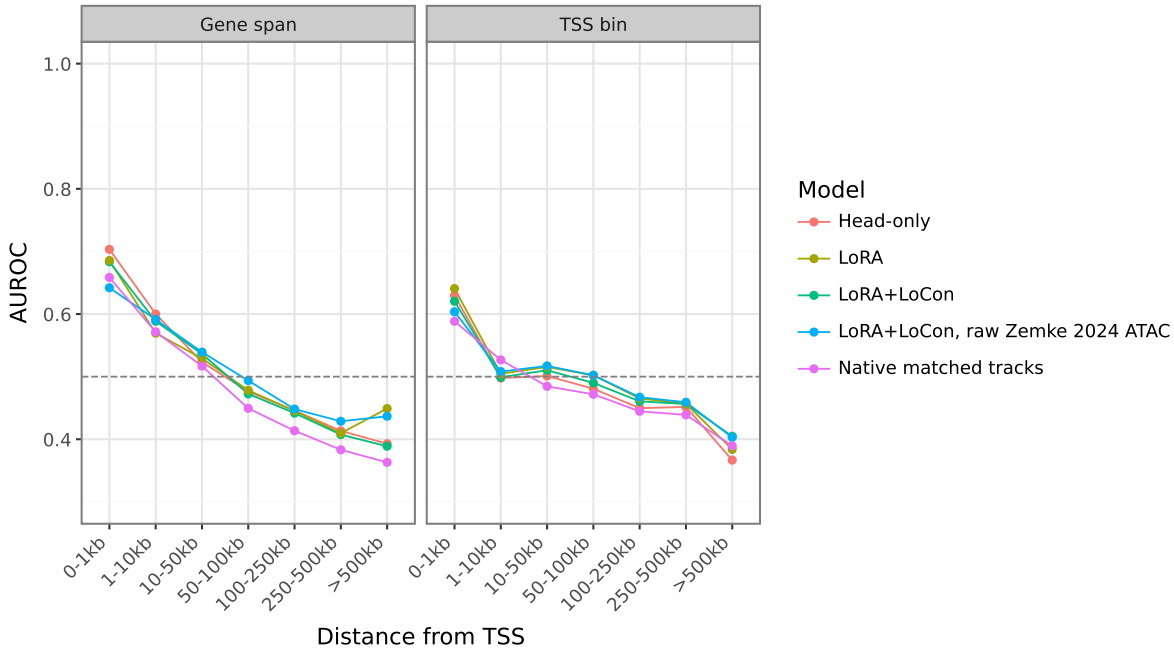


Figure 1: AUROC by distance from the TSS for TSS-bin and gene-span mutation effects. The dashed line is chance performance. Native matched tracks use the biologically defensible panel-specific selections in Table 5.

reference genome. Absolute effects are scored separately for each biosample against the positive and negative variants in each panel group; strand and channel tracks are grouped by source, biosample, and assay. Ranks are among the 371 RNA-seq or 273 CAGE biosample groups. The full rank distribution is shown in Figure 4.

Table 4 lists the highest-AP native track for each cell type after excluding the four biosample/assay groups labeled Testis, leaving 640 groups. Winners are selected separately for the TSS-bin and gene-span summaries across all remaining RNA-seq and CAGE contexts. Several are still non-neural tissues, so these maxima do not establish cell specificity. The heatmap below instead focuses on annotated direct or plausible cell-type candidates.

Table 4: Highest average-precision native RNA-seq or CAGE track for each eQTL cell type after excluding all four biosample/assay groups labeled Testis. Winners are selected independently by summary from the 640 remaining groups.

eQTL cell type	Best TSS-bin track	AP	Best gene-span track	AP
Ast	Dorsolateral prefrontal cortex (ENCODE, total RNA-seq)	0.400	Cerebellum (ENCODE, total RNA-seq)	0.454
End	Tibial nerve (ENCODE, total RNA-seq)	0.259	Heart (ENCODE, total RNA-seq)	0.286
Ext	Dorsolateral prefrontal cortex (ENCODE, total RNA-seq)	0.474	Body of pancreas (ENCODE, total RNA-seq)	0.530
IN	Dorsolateral prefrontal cortex (ENCODE, total RNA-seq)	0.405	Metanephros (ENCODE, total RNA-seq)	0.462
MG	Spleen (ENCODE, total RNA-seq)	0.345	Spleen (ENCODE, total RNA-seq)	0.418
OD	Dorsolateral prefrontal cortex (ENCODE, total RNA-seq)	0.567	Dorsolateral prefrontal cortex (ENCODE, total RNA-seq)	0.607
OPC	Dorsolateral prefrontal cortex (ENCODE, total RNA-seq)	0.452	Parietal lobe (ENCODE, total RNA-seq)	0.522

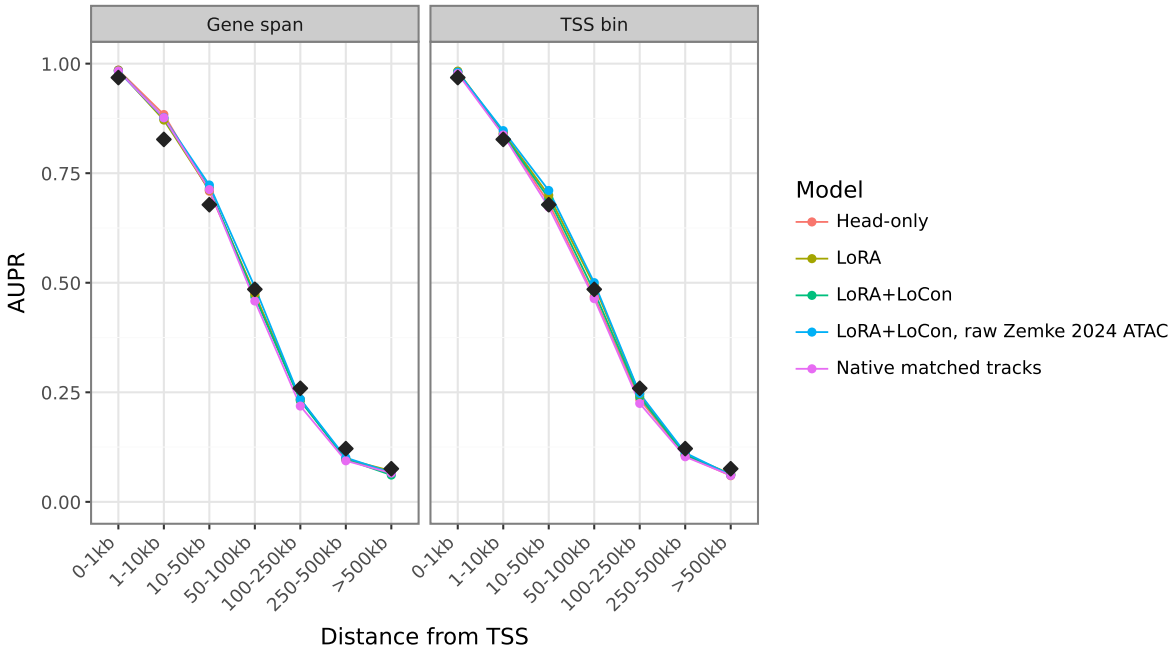


Figure 2: AUPR by distance from the TSS for TSS-bin and gene-span mutation effects. Black diamonds show the random-ranking reference, equal to the positive variant fraction in each distance bin. Native matched tracks use the biologically defensible panel-specific selections in Table 5.

Spleen's lead for both microglia eQTL summaries is biologically plausible given its macrophage content, but a bulk spleen signal is not a microglia-specific measurement. The non-testis winners for other panels remain heterogeneous, including tibial nerve, heart, pancreas, and metanephros.

For the native matched-track curves, one eligible track was selected per cell type as the candidate with the highest mean AP across TSS-bin and gene-span summaries. Eligibility requires a direct cell label or an explicit biological proxy. Inhibitory-panel candidates exclude glutamatergic, motor-neuron, and neuronal-stem-cell tracks; spleen is included as a macrophage-rich tissue proxy for microglia. Mature oligodendrocyte channels are unavailable, so a brain-region track is used. The resulting panel-specific matches and their selection APs are shown in Table 5. These tracks were selected and evaluated on the same variants, so their results are descriptive and may be optimistic.

Table 5: Biologically defensible native tracks used for the matched-track distance curves. Tracks were selected independently per cell type by mean AP across the two summaries.

Cell type	Selected native track	Match type	TSS AP	Gene AP
Ast	Astrocyte (ENCODE total RNA-seq)	Direct cell label	0.335	0.341
End	Endothelial cell (ENCODE total RNA-seq)	Direct cell label	0.226	0.242
Ext	Glutamatergic neuron (ENCODE total RNA-seq)	Related neuronal cell	0.440	0.496
IN	Neuron (FANTOM hCAGE)	Generic neuronal proxy	0.358	0.348
MG	Spleen (ENCODE total RNA-seq)	Myeloid-rich tissue proxy	0.345	0.418
OD	Dorsolateral prefrontal cortex (ENCODE total RNA-seq)	Bulk brain proxy	0.567	0.607
OPC	Dorsolateral prefrontal cortex (ENCODE total RNA-seq)	Bulk brain proxy	0.452	0.507

Monocyte tracks are the clearest microglia proxy. On MG panels, generic monocyte CAGE ranks first of 273 biosample groups for TSS AP (0.320) and second for gene-span AP (0.308), versus

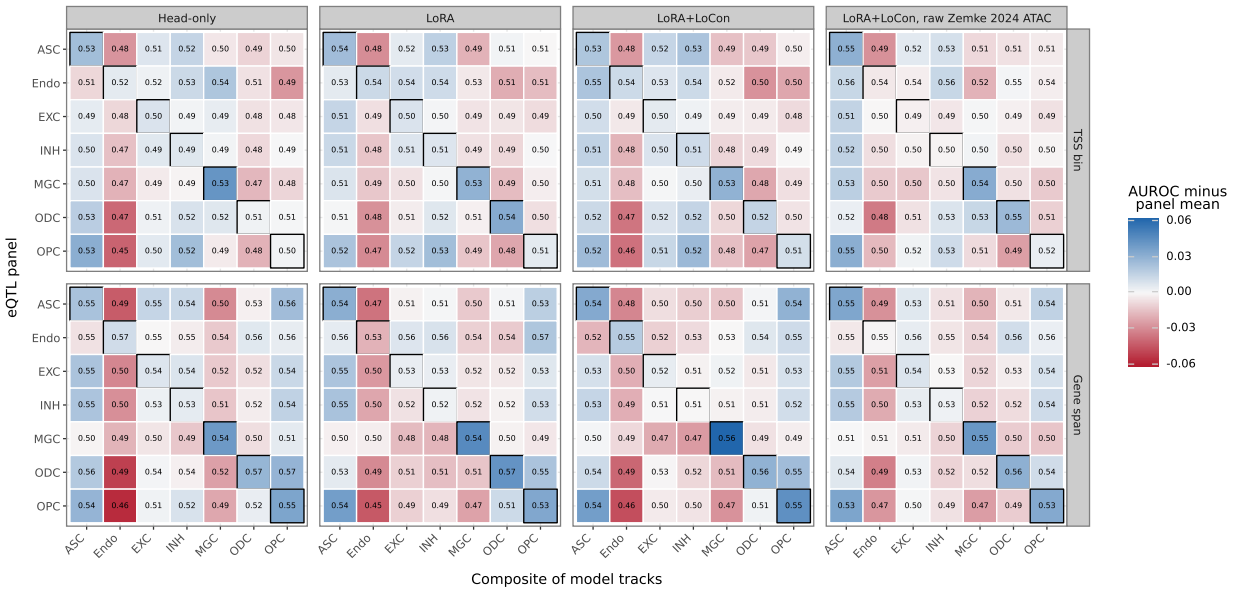


Figure 3: AUROC of each panel group (rows) scored with each track-group composite (columns). Color shows AUROC minus the row mean; black outlines mark the matched composite.

unmatched-track medians of 0.261 and 0.242. The CD14-positive, CD16-negative classical monocyte CAGE track ranks seventh for TSS AP (0.306) and fourth for gene-span AP (0.295). CD14-positive monocyte RNA-seq ranks 27th of 371 for TSS AP (0.307) and 47th for gene-span AP (0.332). This supports CD14-positive monocytes as a useful prediction proxy for these eQTL panels, without equating monocytes with microglia biologically.

There is no mature oligodendrocyte-labeled native RNA-seq or CAGE channel. The single oligodendrocyte-precursor CAGE channel ranks 226th of 273 on OPC TSS AP (0.354) and 246th on gene-span AP (0.336), below the unmatched-track medians (0.362 and 0.345). Bulk brain tracks perform better: dorsolateral prefrontal cortex RNA-seq ranks first of 371 for OD panels on both summaries (AP 0.567 and 0.607), and brain-region tracks also rank highly for OPC panels. For OPC panels, dorsolateral prefrontal cortex RNA-seq ranks first for TSS AP and third for gene-span AP. These are broad neural-tissue proxies, not evidence of oligodendrocyte-specific prediction. Astrocyte tracks are weakly ranked; endothelial tracks are moderately enriched over unmatched tracks. The available glutamatergic and generic neuronal tracks provide mixed results, and there is no inhibitory-neuron-specific native channel.

Figure 5 shows average precision (AP) for candidate tracks across all seven eQTL cell types. Candidate rows are the union of tracks ranking in the top five annotated candidates for at least one cell type and one summary; AP is then shown for every cell type. Tracks are grouped by output type, biosample, source, and assay. Separate panels retain the TSS-bin and gene-span summaries. Because positive-class prevalence differs between cell-type panels, AP is most directly compared between tracks within the same column.

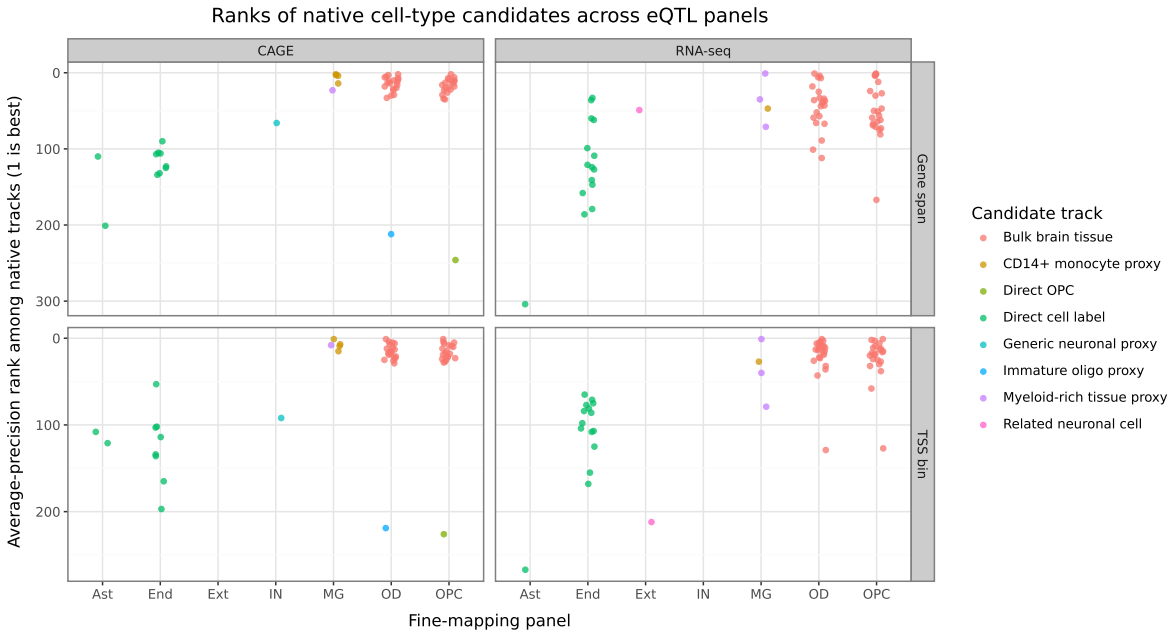


Figure 4: Average-precision rank of native tracks annotated as direct cell labels or plausible proxies within each fine-mapping panel. Rank 1 is best; RNA-seq and CAGE biosample groups are ranked separately. Bulk-brain ranks indicate tissue relevance and should not be interpreted as cell specificity.

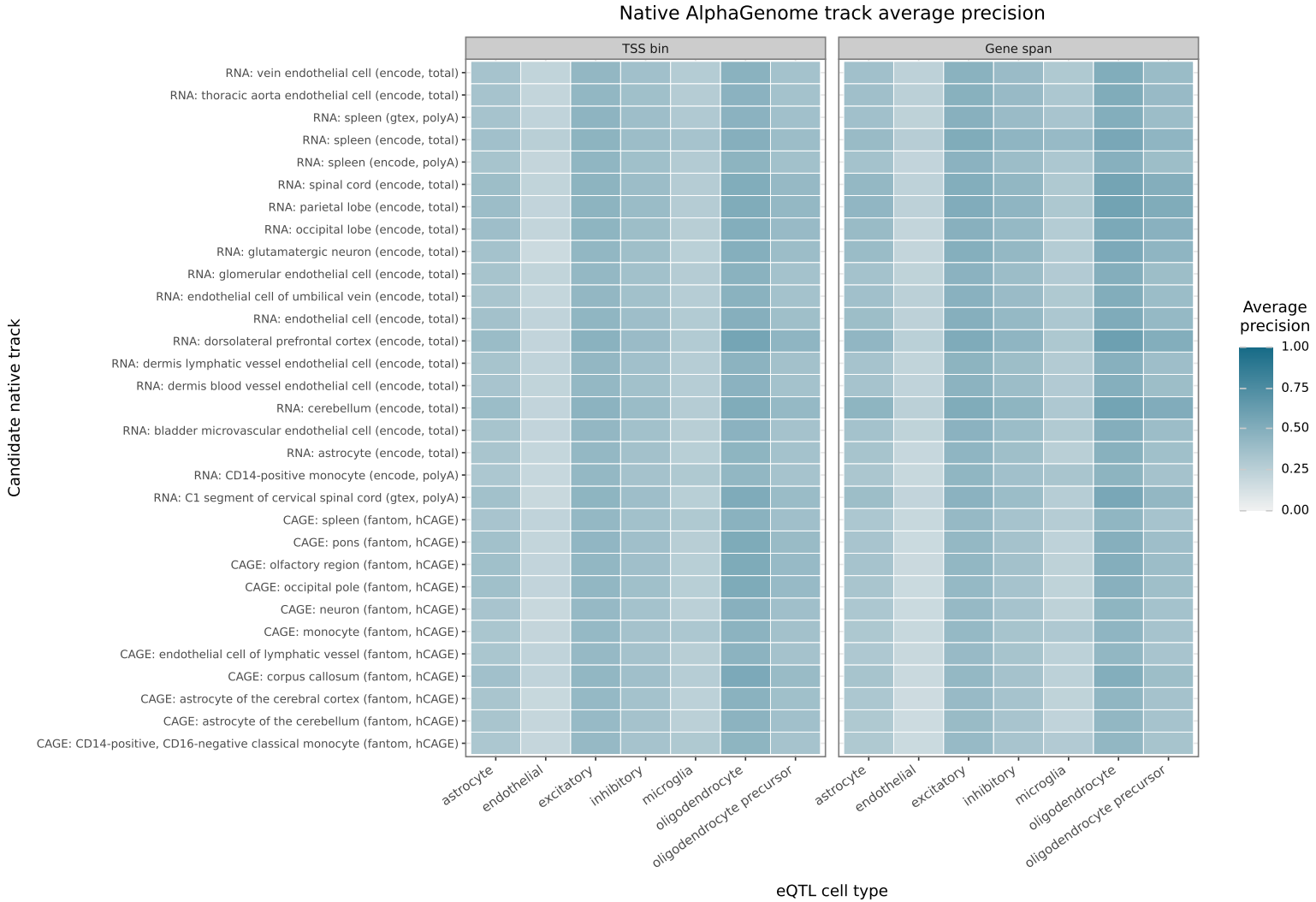


Figure 5: Native AlphaGenome candidate-track average precision by eQTL cell type. Rows are the union of tracks ranked in the top five annotated candidates for at least one cell type and summary; columns show AP for each cell type.

These track ranks are descriptive rather than a formal statistical test of cell-type specificity. Native track availability and assay/source coverage differ across cell classes, and most fine-mapping labels have no exact native counterpart.

5 Implementation details

Fine-mapping tables are streamed from compressed PIP files. Variant alleles are checked against the reference genome before constructing the alternate one-hot sequence. The same reference and alternate sequences are evaluated for every model. The head-only checkpoint was trained on the target tracks without LoRA or LoCon adapters; the fine-tuned models retain trained heads and adapters. All model prediction calls use the custom forward path so the fine-tuned heads, rather than the delegated pretrained variant scorer, are evaluated. Native AlphaGenome variant effect predictions (VEPs) use the original model's valid RNA-seq and CAGE tracks, with predictions reduced on device to the TSS bin and gene span before signed alternate-minus-reference effects are stored. The full matrices and channel/variant metadata accompany the analysis outputs. Candidate tracks were ranked against all biosample groups separately for each panel and summary; the plotted annotations distinguish direct labels, neuronal/monocyte proxies, oligodendrocyte-precursor tracks, and bulk brain tissue.